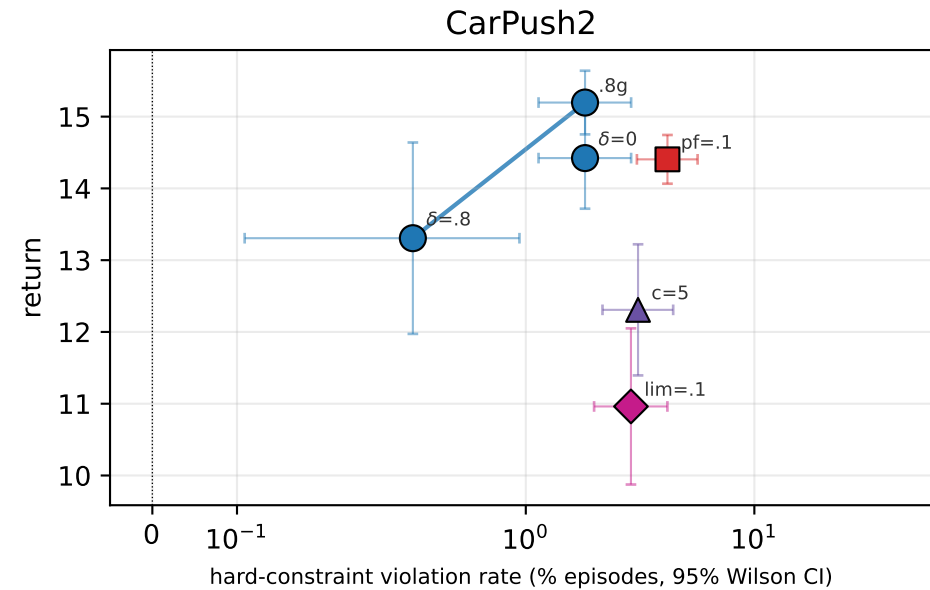
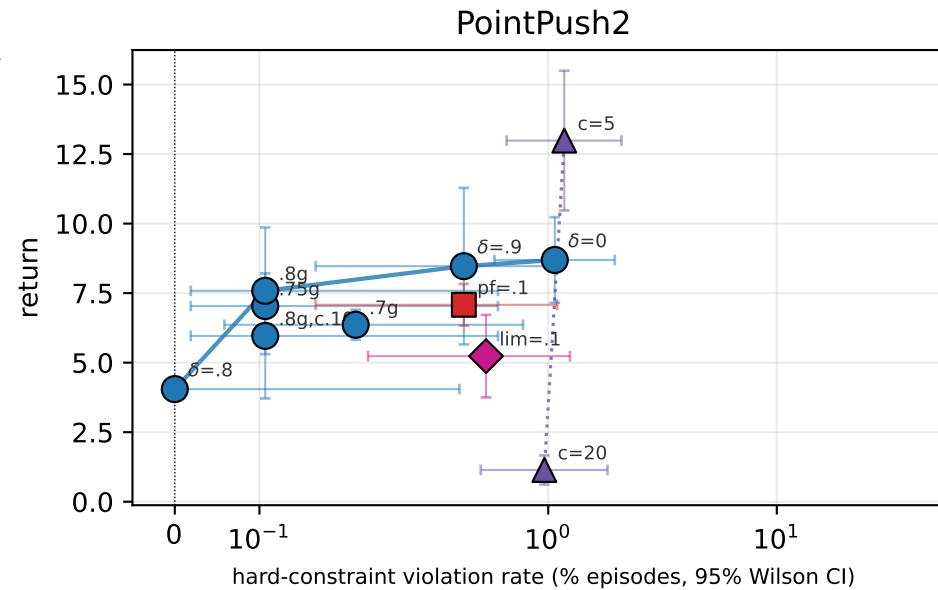


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



● SafeQT-typed (ours)

SafeOT-typed (ours)

- Safety typed (Gals)
- FDBL (constraint based)

FDPI (constraint-based)

◆ SAC-Lagrangian (penalty-based)

SDAC (multi-constraint SOTA)

SDAC (multi-constraint SOTA)

- SRCPO (multi-constraint SOTA)

— typed Pareto frontier ($\text{soft} \leq 2$)

TA) SAC-Pen coefficient sweep

EDBL n. sweep

--- FDPI p_f sweep

--- SAC-Lag cost-limit sweep